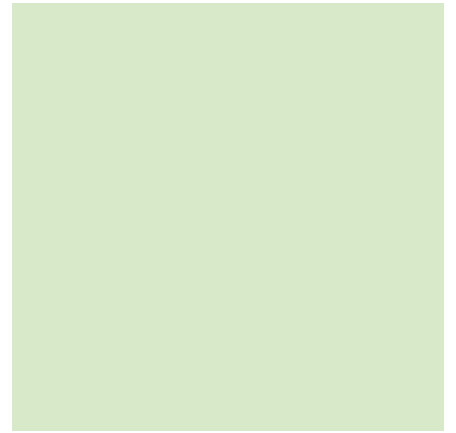
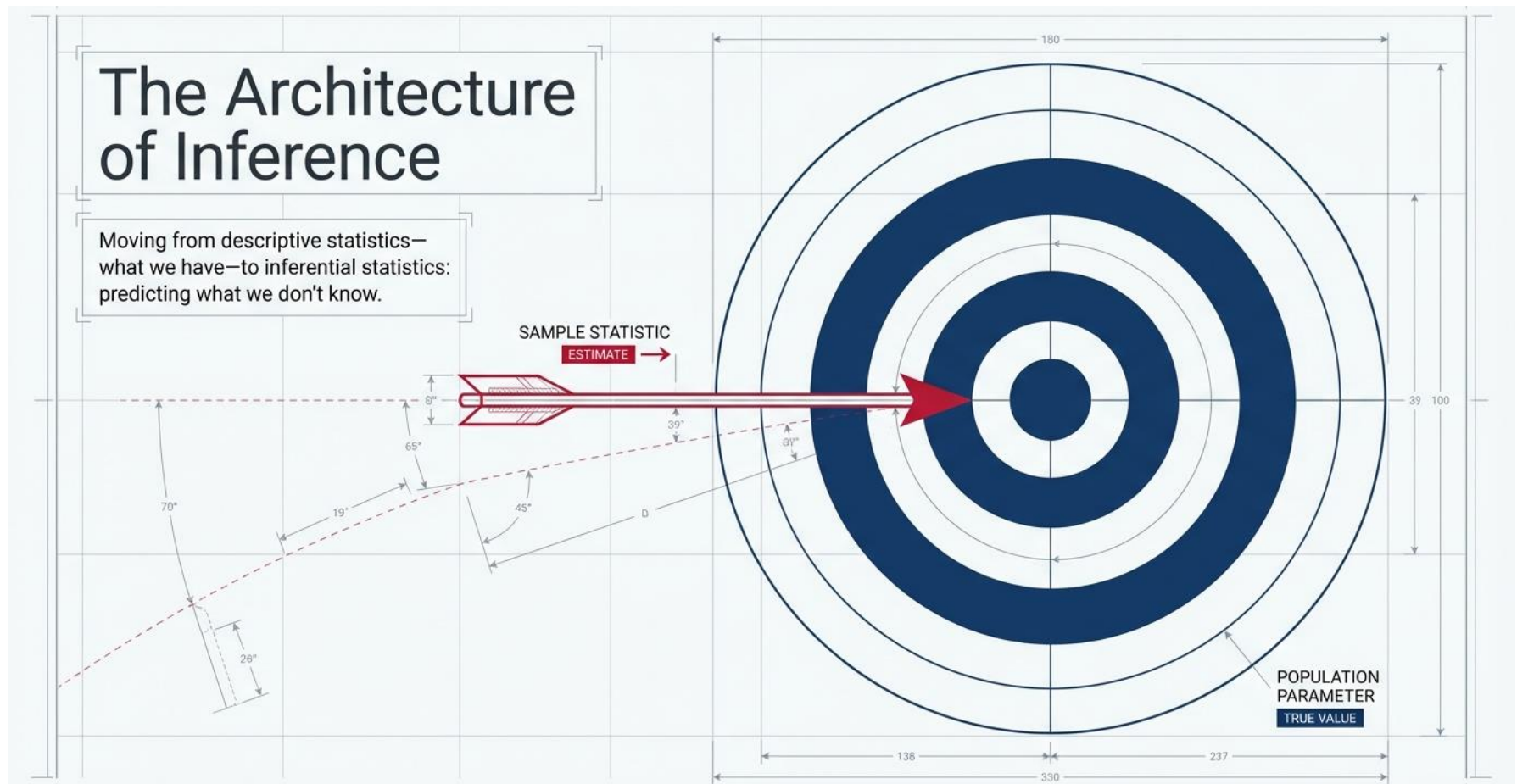




Sampling Distributions

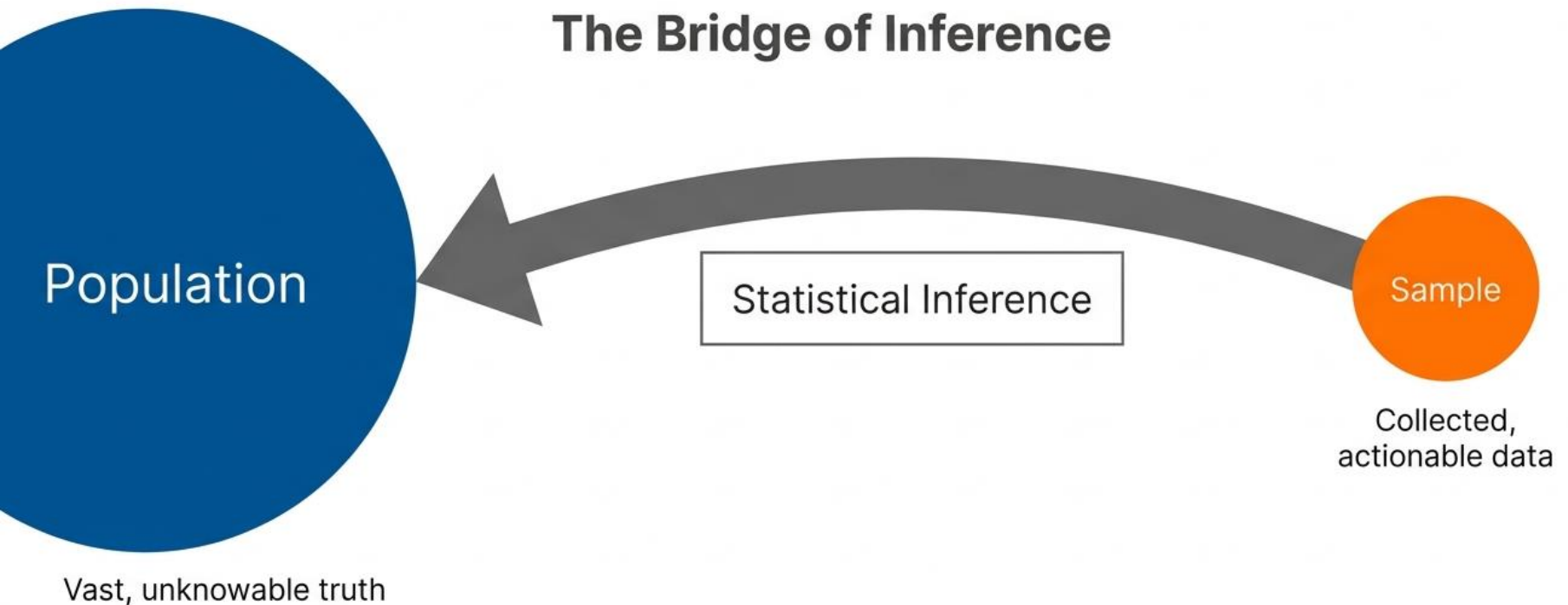
+ A Sampling Distribution

Inferential statistics allow the researcher to come to conclusions about a population on the basis of descriptive statistics about a sample.



Statistical inference spans the gap between known data and unknowable populations.

We cannot examine an entire population. Instead, we collect data from a representative sample and use it to draw broader conclusions.

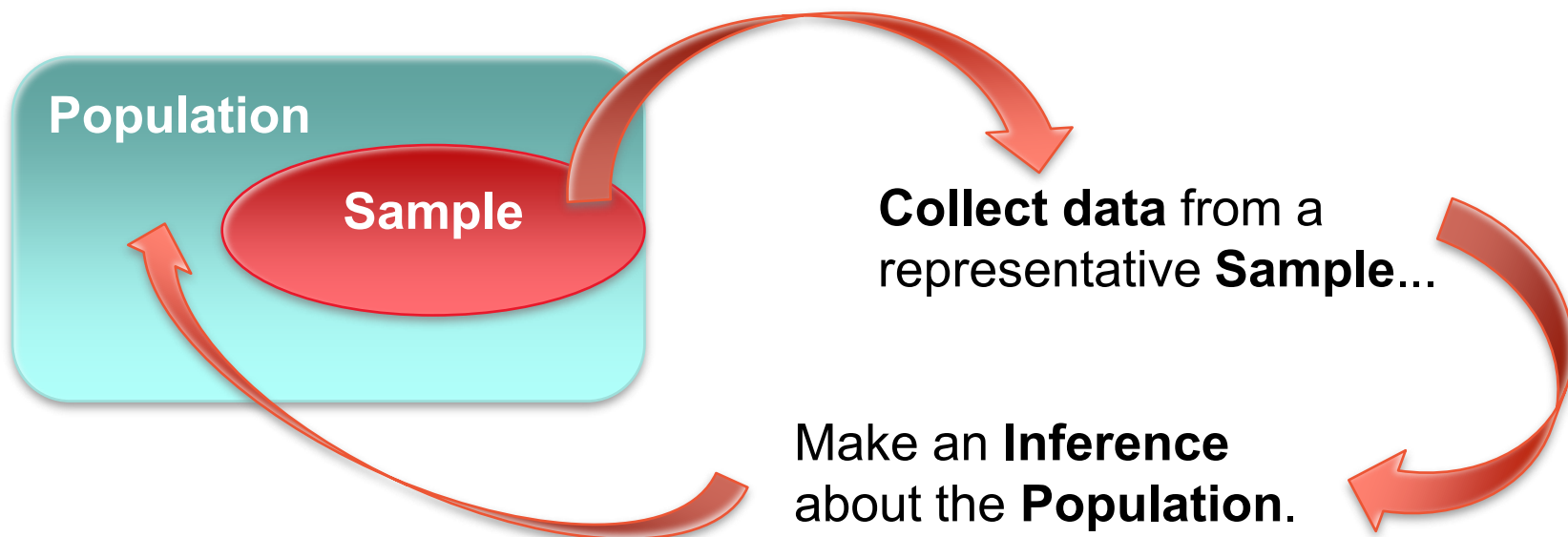


Introduction

The process of *statistical inference* involves using information from a sample to draw conclusions about a wider population.

Different random samples yield different statistics. We need to be able to describe the *sampling distribution* of possible statistic values in order to perform statistical inference.

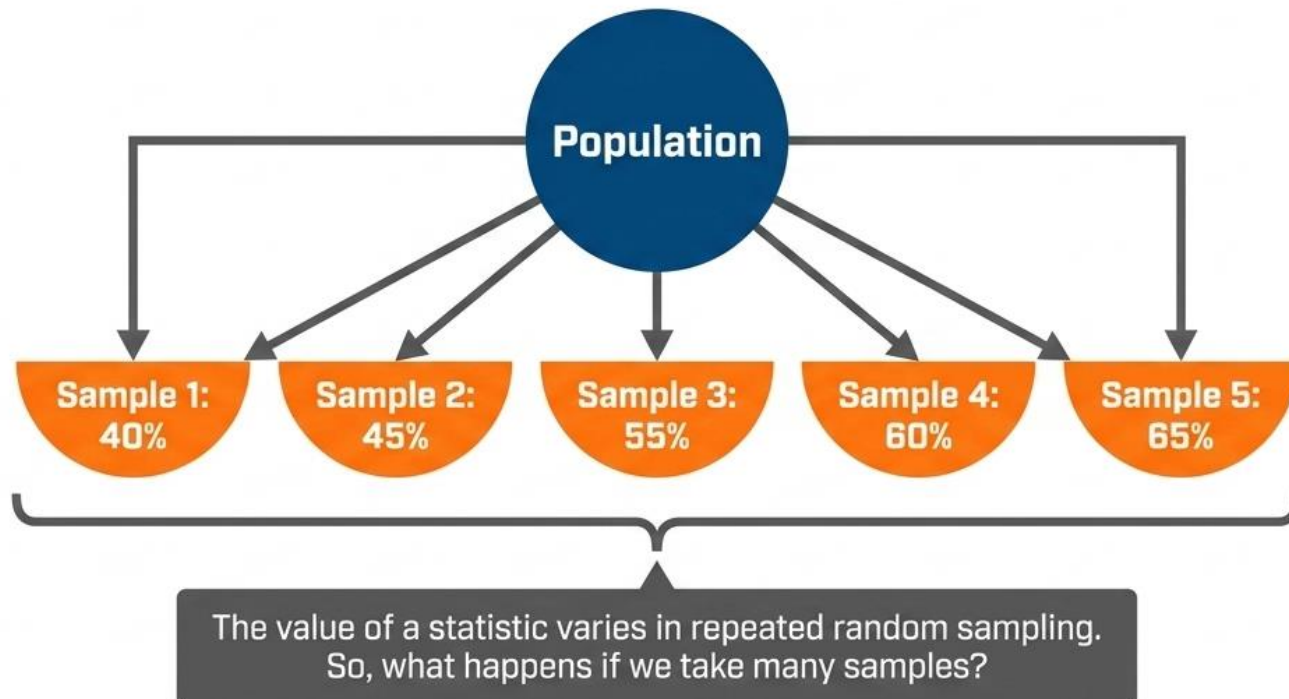
We can think of a statistic as a random variable because it takes numerical values that describe the outcomes of the random sampling process. Therefore, we can examine its probability distribution.



Every random sample yields a different statistic.

Because a statistic takes numerical values that describe outcomes of a random process, it acts as a random variable. This basic fact is called **sampling variability**.

The Variability Multiplier



Parameters belong to populations; statistics belong to samples.

As we use data to draw conclusions, we must be absolutely clear about whether a number describes our small subset or the wider whole.

The Actors Matrix

		The Source	The Measurement
The Source	Measurement	Population (The entire group of interest)	Parameter (A number describing a population characteristic. Usually unknown.)
		Sample (The representative subset)	Statistic (A number computed directly from sample data. Used to estimate the parameter.)

Mnemonic Anchor

p ————— **p** →

**Parameters come
from Populations**

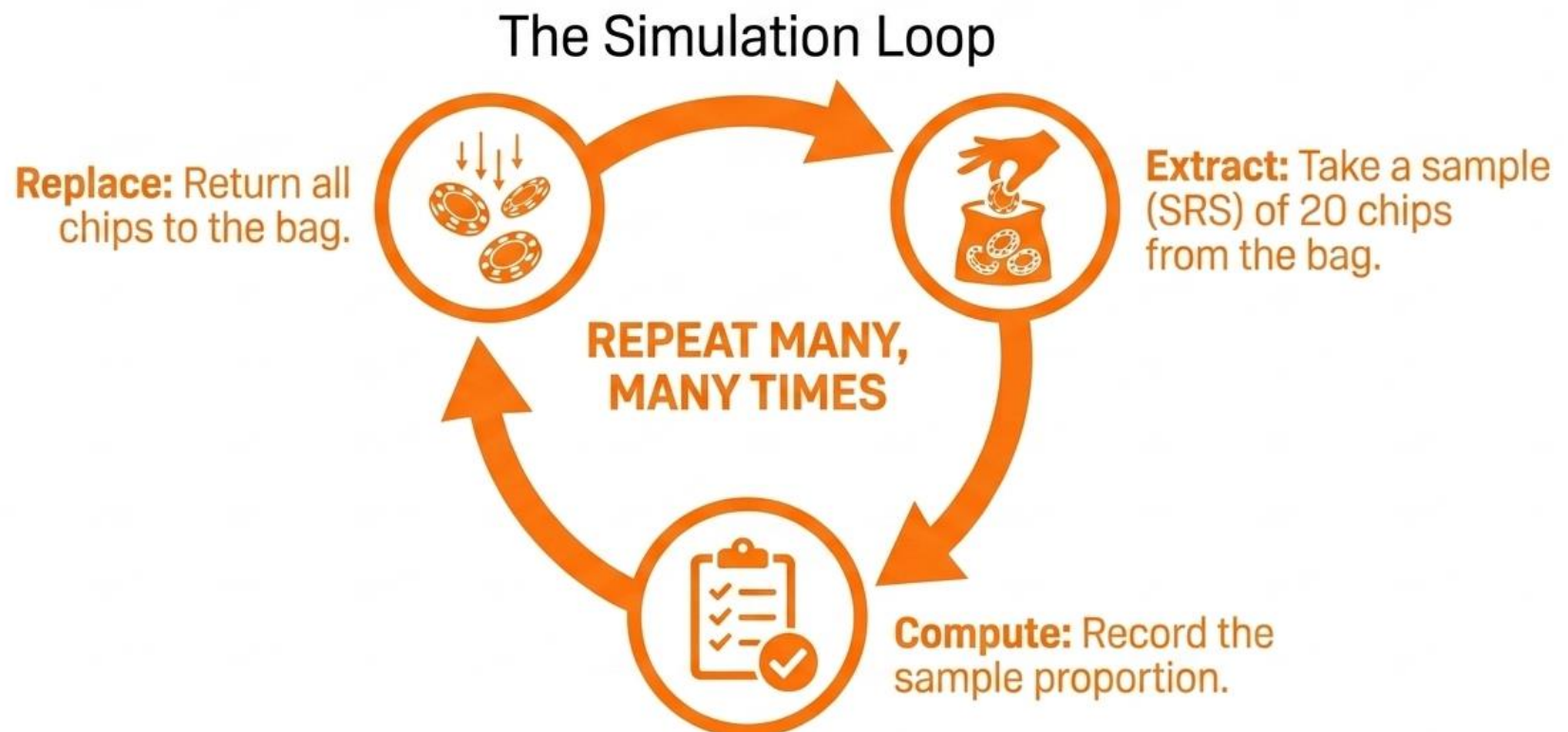
S ————— **S** →

**Statistics come
from Samples**

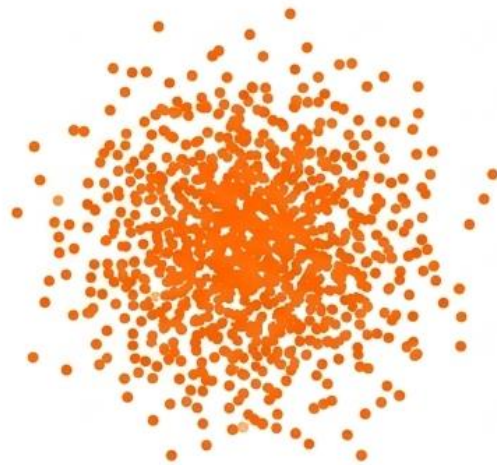


Simulating repeated samples requires an identical, cyclical process.

Consider a population of 200 chips. In practice, obtaining the true mathematical distribution is difficult. Instead, we use simulation to imitate the process of taking many, many random samples (with replacement).

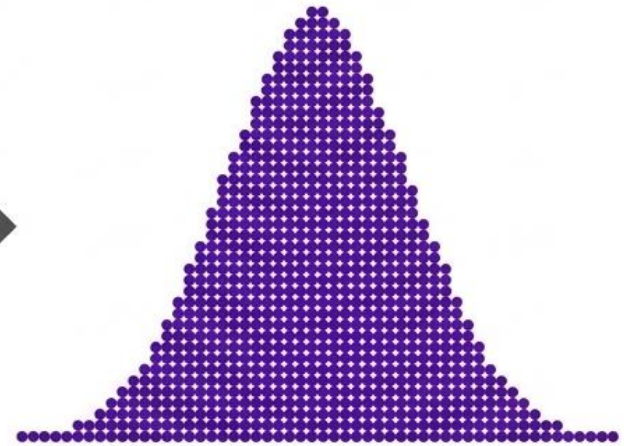


The sampling distribution maps all possible samples of the same size.



All possible sample statistics

Aggregated
and graphed

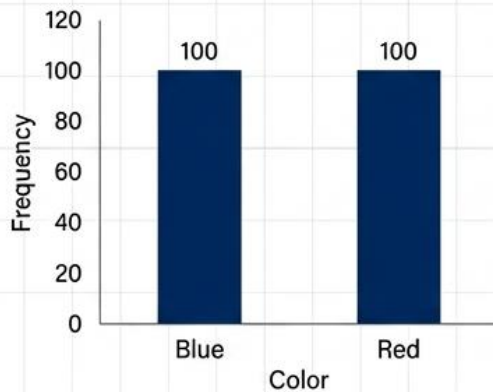


Theoretical sampling distribution

Definition: The sampling distribution of a statistic is the theoretical distribution of values taken by the statistic in all possible samples of the same size from the same population.

The Three Distinct Distributions

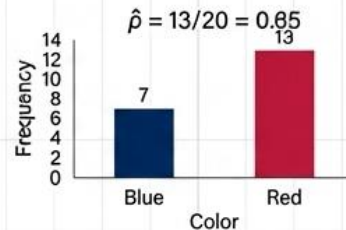
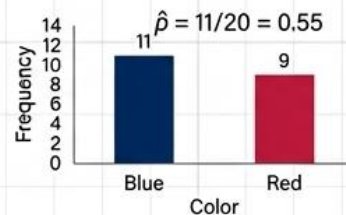
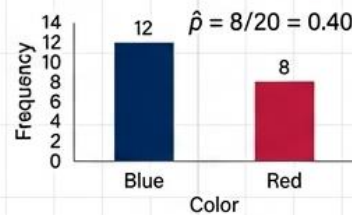
Population Distribution



(All individuals)

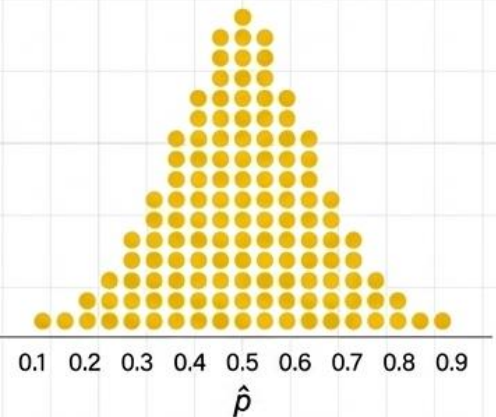
Parameter: $p = 0.50$

Distributions of Sample Data



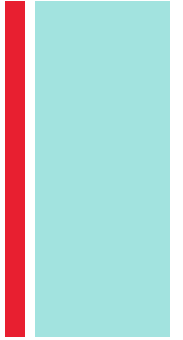
(Individuals in one sample)

Sampling Distribution



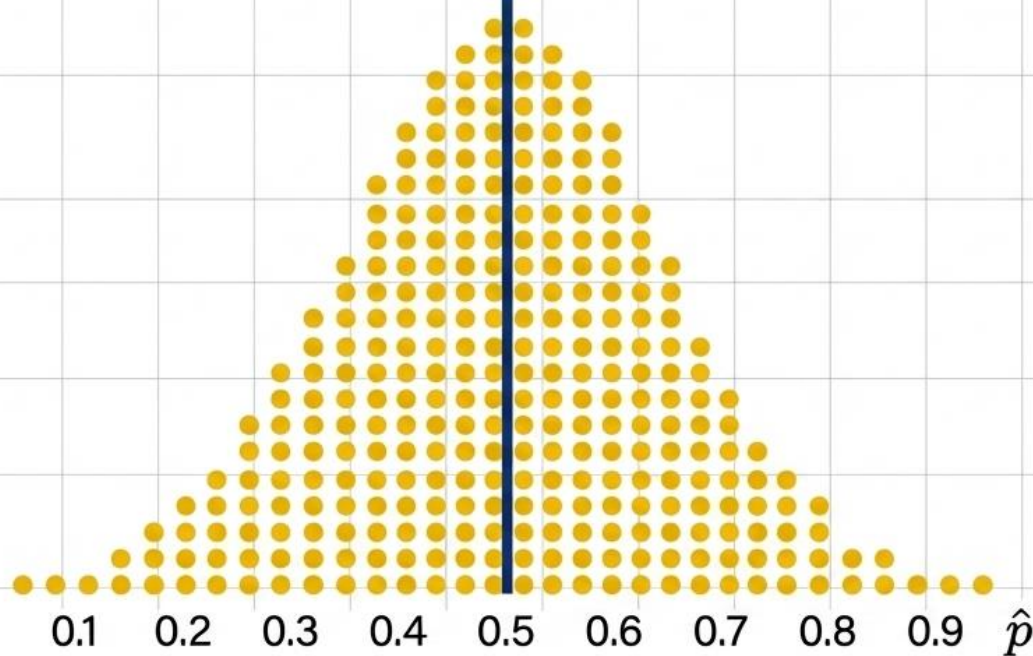
(The statistic values from all possible samples)

+ Sampling Distribution



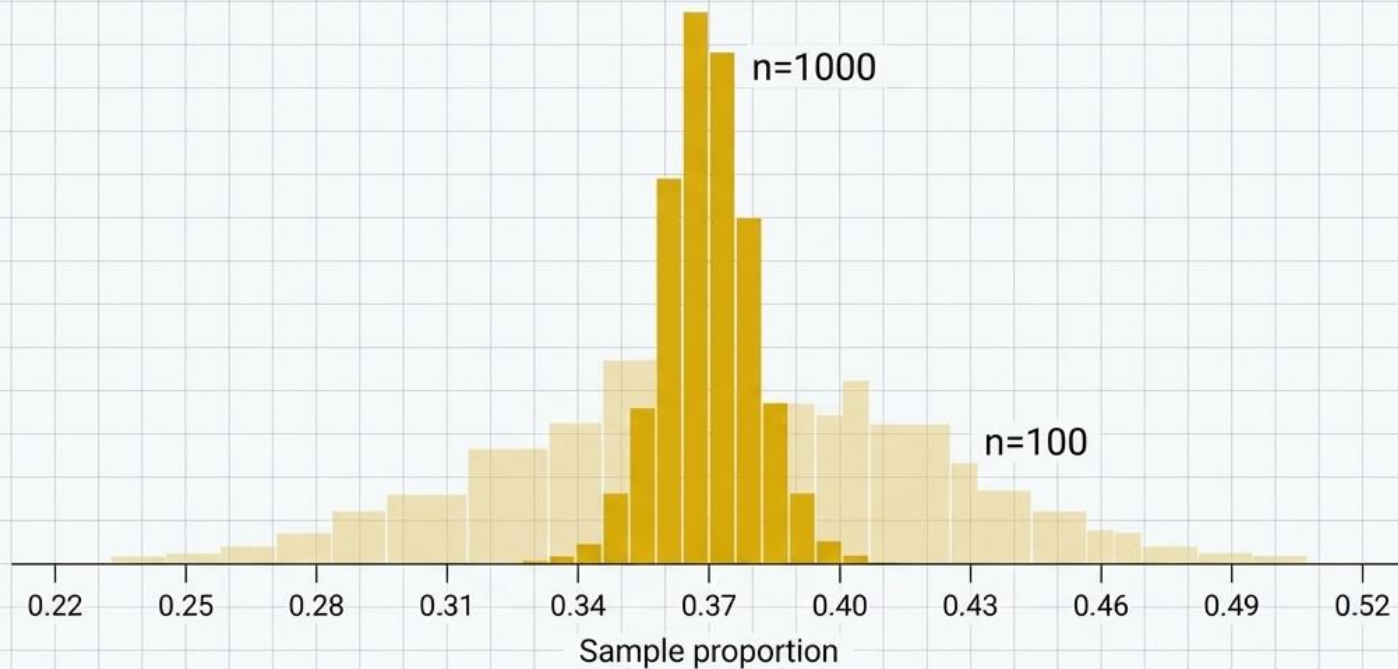
- A theoretical frequency distribution of the scores for or values of a statistic, such as a mean. Any statistic that can be computed for a sample has a sampling distribution.
- A sampling distribution is the distribution of statistics that *would be* produced in repeated random sampling (with replacement) from the same population.
- It is all possible values of a statistic and their probabilities of occurring for a sample of a particular size.
- Sampling distributions are used to calculate the probability that sample statistics could have occurred by chance and thus to decide whether something that is true of a sample statistic is also likely to be true of a population parameter.

Evaluating Trust: The Center



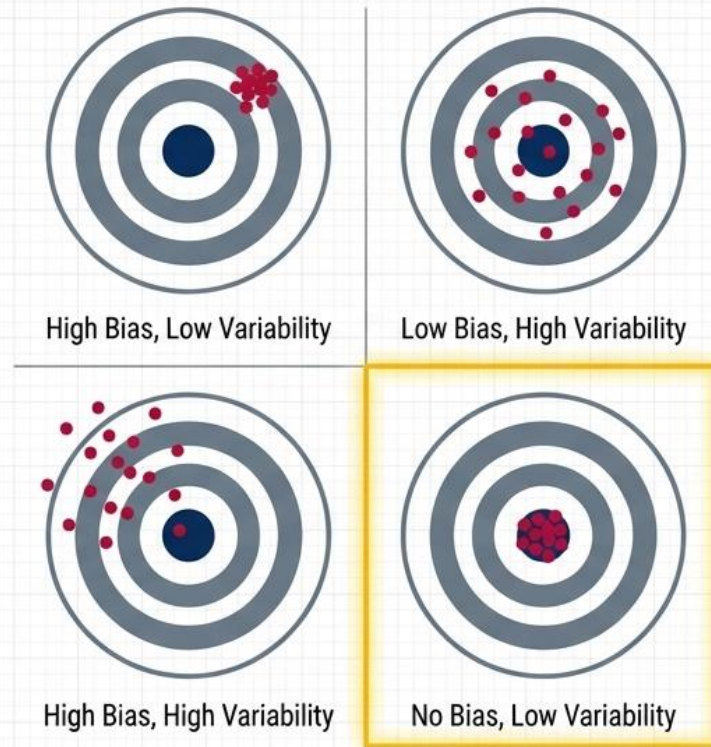
An **unbiased estimator** means our aim is true. The mean of the sampling distribution is exactly equal to the true value of the population parameter being estimated.

Evaluating Trust: The Spread



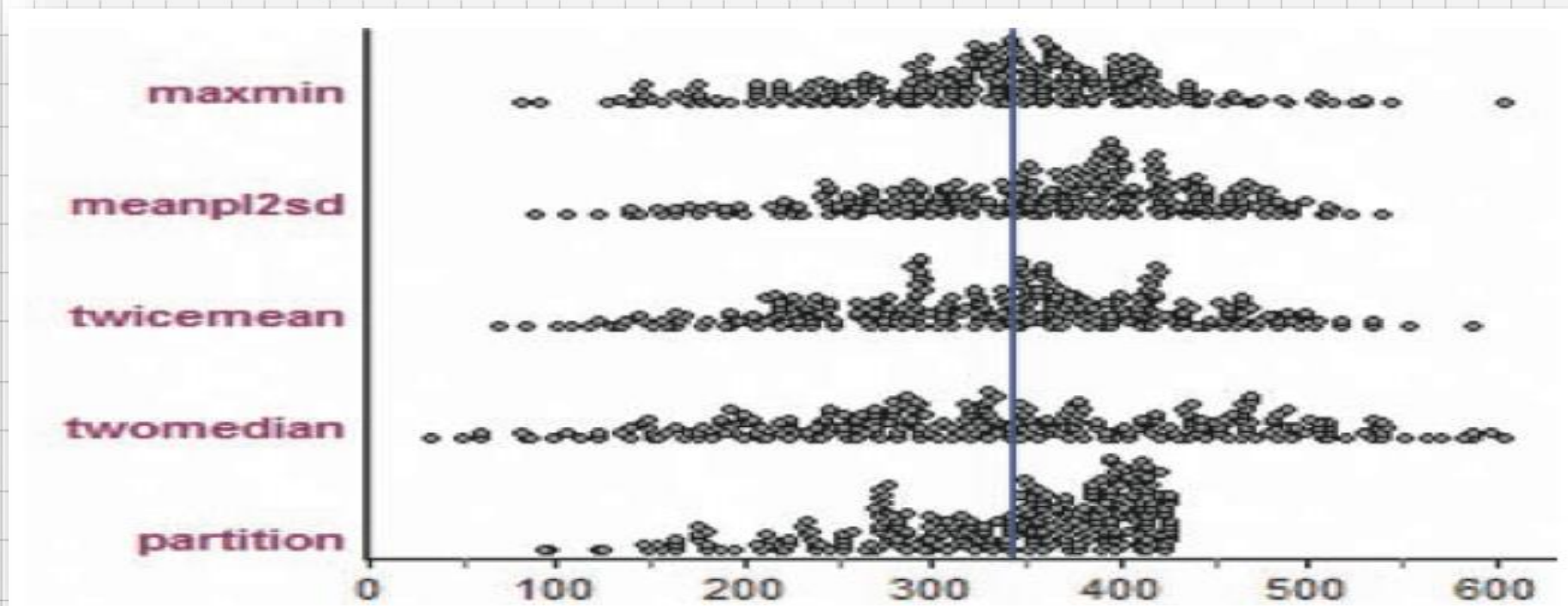
Low variability is better. Larger samples have a clear advantage: they drastically reduce the spread of the sampling distribution, making it much more likely to produce an estimate close to the true parameter.

The Diagnostic Matrix



To get a trustworthy estimate, we must use a statistic that provides the ideal condition: no bias and minimum variability.

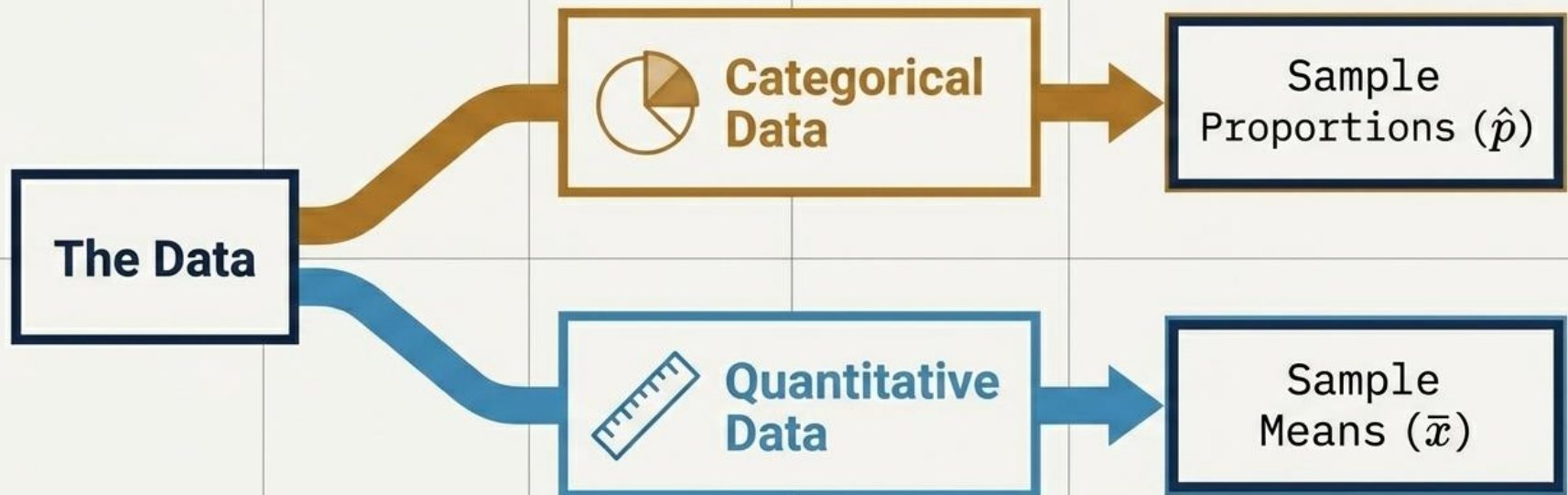
Evaluating Trust: The Shape



Sampling distributions can take on many shapes depending on the population and the statistic chosen. The shape determines how we calculate inference.

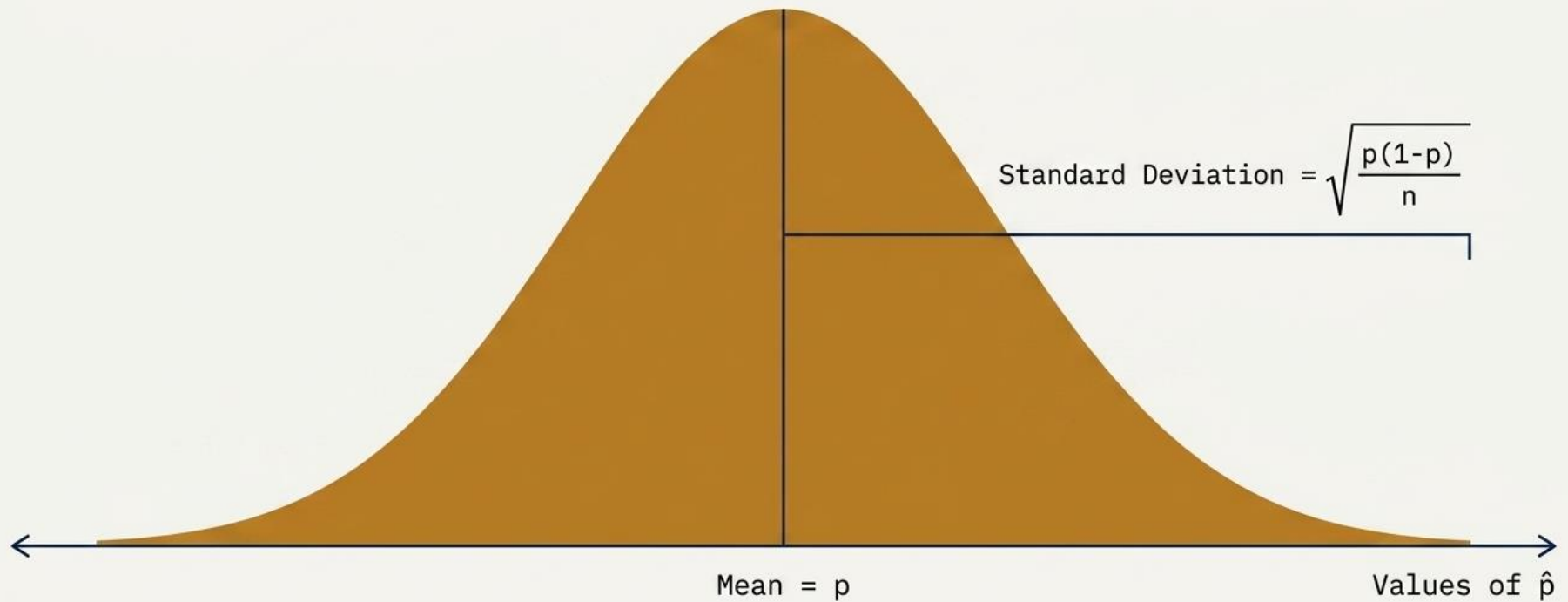
Two paths to the Normal distribution.

The rules of sampling depend entirely on what you are measuring. We must separate our tools based on whether we are asking "What percentage?" (Proportions) or "What is the average?" (Means).



Branch 1: The Sampling Distribution of a Sample Proportion ($\hat{p}$)

Sample proportions arise most often when we are interested in categorical variables (e.g., success/failure). The distribution centers perfectly on the true population proportion (p).



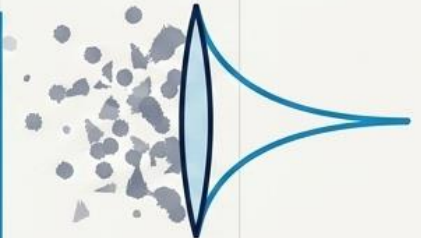
Anatomy of the Spread

Math confirms intuition:
larger sample sizes
yield less variable,
more precise estimates.

$$\sqrt{\frac{p(1-p)}{n}}$$

The inherent variance in the population.
Maximum when $p = 0.5$.

The Focusing Lens.
Because sample size
(n) is in the
denominator, as your
sample grows, the
total spread shrinks.



The Pre-Flight Checklist for Proportions

The Normal Condition

Before you perform Normal calculations, you must ensure the sample is large enough to expect at least 10 successes and 10 failures.

✓ $np \geq 10$

✓ $n(1 - p) \geq 10$

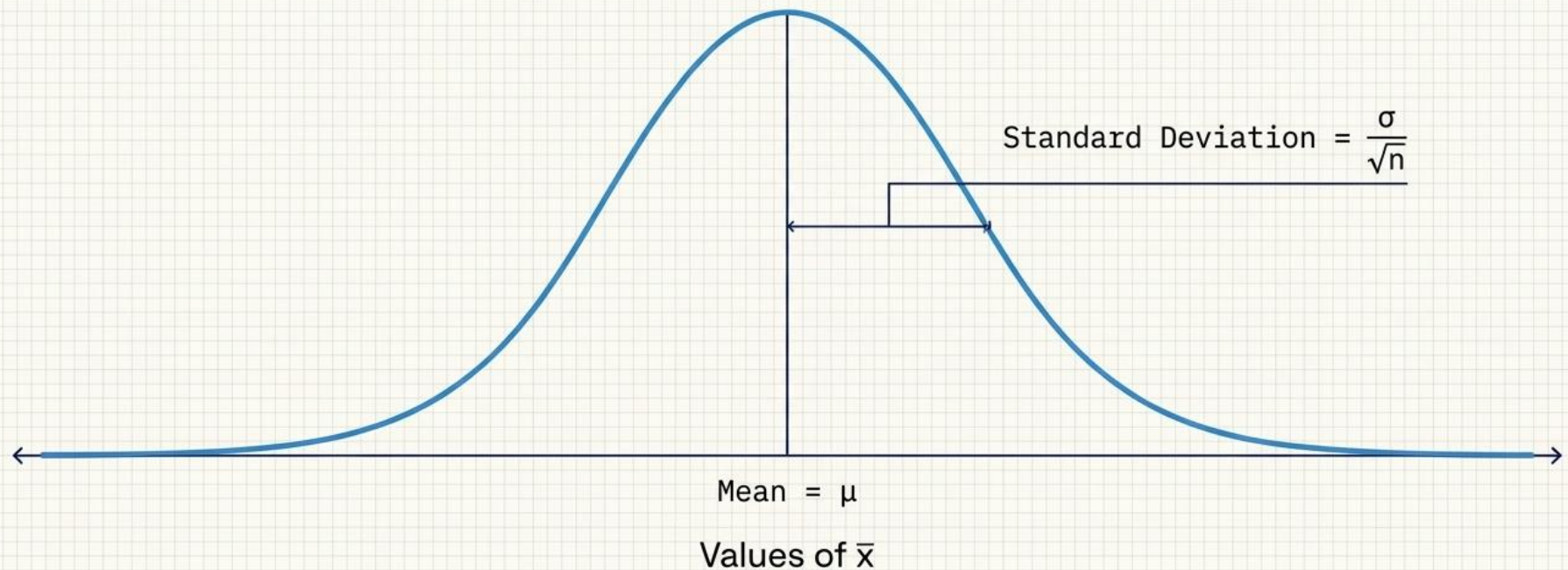
The 10% Condition

To assume independence when sampling without replacement, the sample must be no larger than 10% of the population.

✓ $n \leq 1/10 N$

Branch 2: The Sampling Distribution of a Sample Mean ($\bar{x}$)

When we record quantitative variables, we track statistics like the mean. The sampling distribution of the sample mean is centered precisely at the population mean (μ).



Anatomy of the Spread (Means) .

The sampling distribution is always less spread out than the raw population distribution.

The original population's standard deviation. The raw chaos.

$$\frac{\sigma}{\sqrt{n}}$$

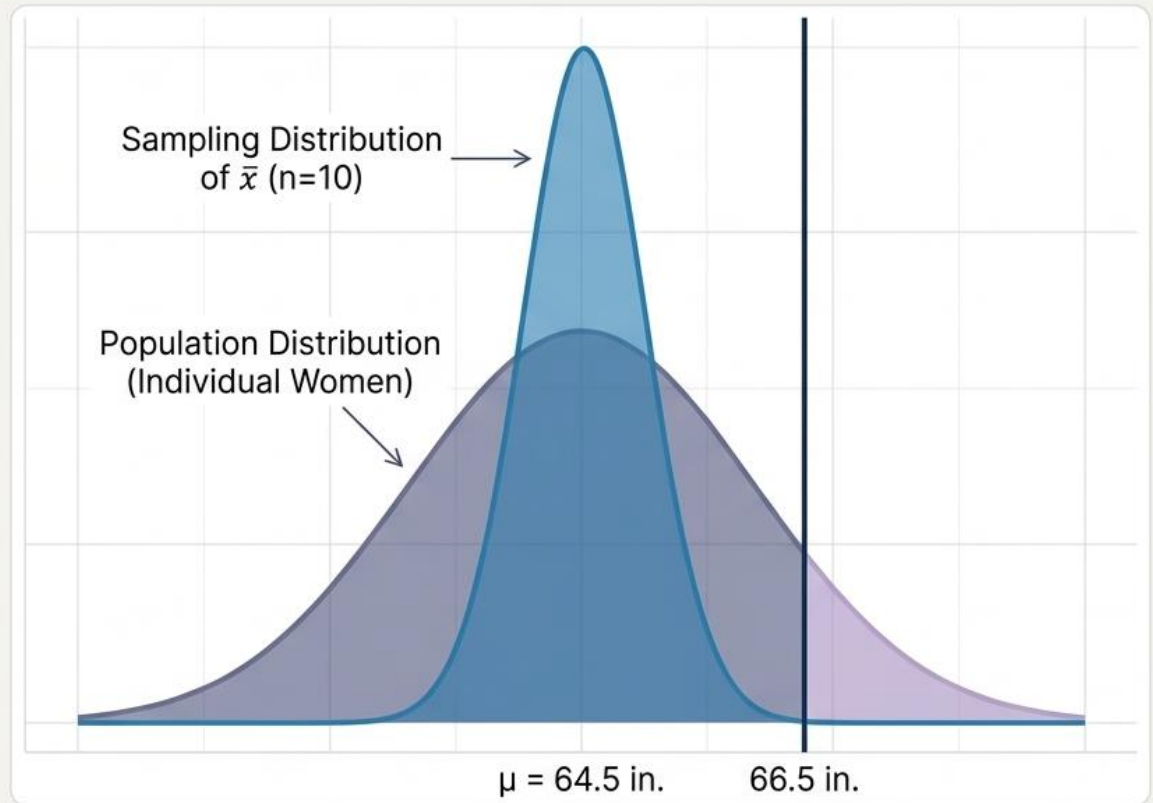
The Focusing Lens. The spread of the sample mean decreases by the square root of your sample size.

Case Study: The Shrinking Shadow of Variance

The height of young women follows a Normal distribution ($N(64.5, 2.5)$).

There is a 21% chance a randomly selected individual is taller than 66.5 inches.

However, the probability that the average height of an SRS of 10 women exceeds 66.5 inches shrinks to less than 1% ($N(64.5, 0.79)$). The extremes average out.



The Diagnostic Matrix

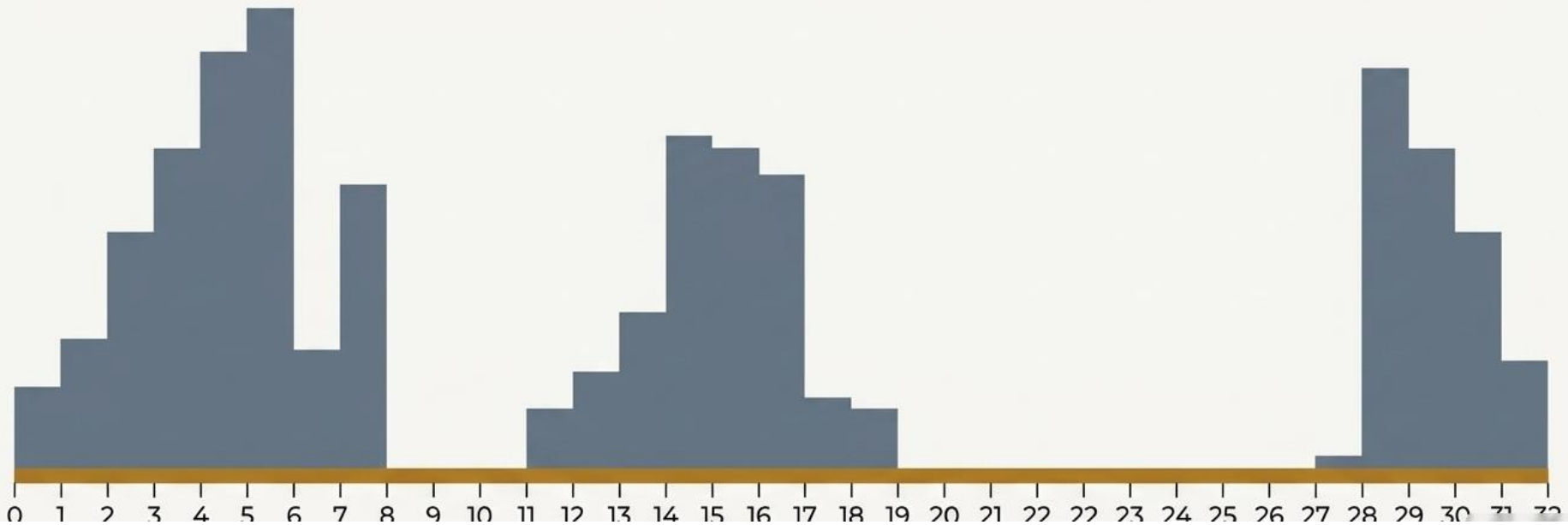
Determine your data type, then follow the corresponding mathematical rules.

	Sample Proportions ($\hat{p}$)	Sample Means ($\bar{x}$)
Data Type	Categorical	Quantitative
Center	Mean = p	Mean = μ
Spread	$\sqrt{\frac{p(1-p)}{n}}$	$\frac{\sigma}{\sqrt{n}}$
Normal Condition	$np \geq 10$ and $n(1-p) \geq 10$	Population is Normal or n is large (CLT)

The Chaos of Reality.

Most real-world population distributions are not strictly Normal. They can be skewed, bimodal, or entirely erratic.

What is the shape of the sampling distribution of sample means when the source population looks like this?



The Central Limit Theorem (CLT)

When the sample is large enough, the sampling distribution is very close to Normal, no matter what shape the population distribution has (provided a finite standard deviation).

